

CosmicCraft AI

AI-Powered Comic Story Creator • Version 1.0

GitHub Repository: <https://github.com/esakki372/cosmiccraft-ai>

Live Demo: <https://cosmiccraft-ai-1.onrender.com>

1. PROJECT OVERVIEW

CosmicCraft AI is a web-based AI-assisted comic creation application. It accepts a creative story idea and processes it through a modular FastAPI backend, Gemini-based content generation, comic-panel processing, layout construction, and PDF export. The application is maintained through GitHub and deployed as a live Render web service.

2. PROBLEM STATEMENT

Creating a comic normally requires separate work for story writing, panel planning, dialogue, visual design, page arrangement, and final export. This can be time-consuming for individual creators and beginners. CosmicCraft AI combines these stages into one guided workflow so a user can begin with a text idea and obtain structured comic content.

3. OBJECTIVES

- Provide a simple browser interface for comic creation.
- Generate comic-oriented outlines and narrative content with Gemini.
- Organize image generation, layout, and export into separate modules.
- Provide FastAPI endpoints for the main application functions.
- Deploy the application as a public Render service.
- Protect API credentials with environment variables.

4. KEY FEATURES

- AI-assisted comic generation from text prompts • Comic outline generation
- Story, narration, and dialogue generation • Character and tone inputs
- Image-generation module for panel visuals • Comic layout construction
- PDF export workflow • Responsive web interface • Health-check endpoint

5. TECHNOLOGY STACK

Frontend & Core

- HTML5 & Browser UI
- Python & FastAPI
- Uvicorn ASGI Server

AI & Processing

- Google Gemini API (Flash/Pro)
- Layout Builder & Exporters
- Git, GitHub, & Render

6. SYSTEM ARCHITECTURE

User Browser → HTML Interface → FastAPI (main.py) → API Routes (app/routes.py) → Gemini generation modules → Image Generation → Layout Builder → PDF Exporter → Comic Preview / Export.

The modular architecture separates web routing, AI generation, visual processing, layout, and export responsibilities.

7. PROJECT STRUCTURE

```
CosmicCraft_AI/  
&|— main.py, requirements.txt, render.yaml, .env.example, README.md  
&|— app/  
  | &|— routes.py, gemini_flash.py, gemini_pro.py  
  | &|— image_generator.py, layout_builder.py, exporters.py  
&|— templates/  
    |— index.html, comic_preview.html, export_success.html
```

8. MODULE DESCRIPTION

- **main.py:** FastAPI application entry point, middleware, templates, static resources, and route registration.
- **app/routes.py:** API operations for comic generation, outline generation, story generation, and health checking.
- **app/gemini_flash.py & pro.py:** Fast and rich Gemini-oriented generation logic.
- **app/image_generator.py & layout_builder.py:** Visual generation and component assembly.
- **app/exporters.py:** PDF export functionality.

9. APPLICATION WORKFLOW & ENDPOINTS

Workflow: 1. User inputs comic idea → 2. Request sent to FastAPI → 3. AI layer creates outline → 4. Narrative & dialogue expanded → 5. Panel visuals prepared → 6. Layout assembled → 7. PDF exported → 8. Result returned to browser.

API Endpoints:

- POST /api/generate-comic — Generates the full comic workflow.
- POST /api/generate-outline — Generates a comic outline.
- POST /api/generate-story — Generates story content from an outline.
- GET /health — Verifies service health.

10. SETUP, SECURITY & DEPLOYMENT

- **Local Setup:** Python 3.9+, clone repo, create virtual env, install requirements via `pip install -r requirements.txt`, configure `.env` with `GEMINI_API_KEY`, and run `uvicorn main:app --reload`.
- **Security:** Secrets are excluded from version control via `.gitignore` and configured directly inside Render service environment variables.
- **Deployment:** Configured via `render.yaml` connected to GitHub. Live demo hosted at <https://cosmiccraft-ai-1.onrender.com>.

11. MILESTONES & CONCLUSION

Milestones: Completed project setup, AI integration, comic processing, web integration, version control consolidation, and Render public deployment.

Conclusion: CosmicCraft AI successfully demonstrates the integration of generative AI with a modular web application for automated comic creation, combining FastAPI, Gemini, and Render deployment.